



Figure 1 Sidebar: Technical Specifications

Project: Analysis: Cerro Dúrika Mass Wasting Event
Location: Talamanca Range, PILA, Costa Rica
Coordinates: Lat 9.3762° Lon -83.3009°

Personnel:

- **Analyst:** Arturo Martín-de-Nicolás (Alumni BSEE/BACS Rice University)
- **Coordination:** Geógrafo José Rivas (Espacio Geográfico, Facebook)
- **Research Associate:** Gemini (AI Model) - Data Synthesis & Structural Modeling

CHRONOLOGY

- **Dec 2015:** Baseline imagery (Fig 2) shows incipient ~200m "V" surface scoring; primary slope remains intact.
- **Apr 2025:** Satellite/Aerial imagery (Fig 1) captures the 220m vertical collapse; event occurs prior to this timestamp.
- **Dec 2025:** Ground-level summiting (J. Salas) via southern/eastern approach; failure remains hidden by northern topography.
- **Mar 2026:** Discovery via remote sensing (A. Martín-de-Nicolás); subsequent drone validation (J. Salas) confirms "ground truth."

DATA PROVENANCE

- **Topography:** 1:50,000 IDECORI Isohyps (SNIT Geoportal)
- **Imagery:** Google Maps/SNIT (April 2025) verified against multi-temporal Google Earth Pro archives (Fig 2)
- **Orientation:** 180° Rotation (Gravity-Aligned / Slope-Down View)
- **Validation:** 2026 Drone Reconnaissance (Johel Salas)

TOPOGRAPHIC BASELINE

- **Source:** 1:50,000 IDECORI Isohyps (SNIT Geoportal)
- **Interval:** 10 meters (Major index at 100m)
- **Status:** Pre-Event Photogrammetric Surface
- **Visualization:** Contours rendered in RED over the failure zone to illustrate the divergence between the historical stable slope and current satellite-observed morphology.

OBSERVATIONS

- **Headscarp:** Massive structural failure initiating at the ~3,120m contour.
- **Displacement:** Approximately 220m vertical wedge collapse (from 3,120m to 2,900m base) channelizing into the headwaters of the Río Dúrika.
- **Secondary Event:** Reactive bank failure observed downstream, likely triggered by lateral erosion from primary debris pulse (Fig 3).

FUTURE RESEARCH

- Volumetric quantification via Structure from Motion (SfM) utilizing recent 4K drone assets.

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